



Running a quick NMAP scan to inventory my network

Use Nmap, the open source network mapper tool, to better understand what's happening in your network.

Posted: March 31, 2020 | | [Valentin Bajrami](#) (Sudoer alumni).



Photo by [Shvets Anna](#) from [Pexels](#)

Nmap, which stands for "Network Mapper," is an open source tool that lets you perform scans on local and remote networks. [Nmap](#) is very powerful when it comes to discovering network protocols, scanning open ports, detecting operating systems running on remote machines, etc. The tool is used by network administrators to inventory network devices, monitor remote host status, save the scan results for later use, and so on.

[Just getting started with networking? Check out the [Linux networking cheat sheet](#).]

The **Nmap** suite includes an advanced graphical user interface and results viewer (**Zenmap**), a flexible data transfer, redirection, and debugging tool (**Ncat**), a utility for comparing scan results (**Ndiff**), and a packet generation and response analysis tool (**Nping**).

Great Linux resources

- [Advanced Linux commands cheat](#)

How we use cookies

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#). By using this website you agree to our use of cookies.



[oad RHEL 9 at no charge](#)
[h the Red Hat Developer](#)
[m](#)
[e to installing applications on](#)

- [Linux system administration skills assessment](#)
- [How well do you know Linux? Take a quiz and get a badge](#)

Why use Nmap?

Besides being free, **Nmap** is very flexible, portable, well-documented, and easy to use. In the following post, we'll walk you through on how to install **Nmap**, use it, and, most important, get more to know about your network.

Installing Nmap

To install **Nmap** on [Red Hat Enterprise Linux 8](#) or [Fedora](#), you'd run:

```
# dnf -y install nmap
```

Substitute **dnf** for **yum** if you are on Red Hat Enterprise Linux 7 or newer. After installing **Nmap**, you can run the **nmap** command without arguments to display all of its options. You also should consult the **Nmap man** page by running **man nmap**.

Using Nmap

Let's assume your local network is 192.168.0.0/24, and you want to run a scan on this network. Running a scan without any argument except the network address yields the following:

```
# nmap 192.168.0.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-06 21:00 CET
Nmap scan report for Archer.lan (192.168.0.1)
Host is up (0.0046s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
1900/tcp  open  upnp
20005/tcp open  btx
MAC Address: 50:ff:BF:ff:ff:AC (Tp-link Technologies)

Nmap scan report for Lyric-1111C2.lan (192.168.0.101)
Host is up (0.013s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: B8:dd:A0:dd:dd:C2 (Resideo)
Multiple networks can be scanned at once. For example
nmap 192.168.0.0/24 10.80.0.0/24
```

Multiple networks can be scanned at once. For example:

```
# nmap 192.168.0.0/24 10.80.0.0/24
```

If we want to run a quick scan of machines in our network without trying to see if any port is open, we run:

```
# nmap -sn 192.168.0.0/24
```

 **How we use cookies**

× g like:

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#).

By using this website you agree to our use of cookies.

```
# nmap -sn 192.168.0.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-06 21:24 CET
Nmap scan report for Archer.lan (192.168.0.1)
Host is up (0.016s latency).
MAC Address: 50:C7:FF:FF:15:FF (Tp-link Technologies)
Nmap scan report for Lyric-1111C2.lan (192.168.0.101)
Host is up (0.96s latency).
MAC Address: B8:FF:FF:11:FF:C2 (Resideo)
MAC Address: 88:DD:EA:DD:CE:37 (Texas Instruments)
Nmap scan report for SoundTouch-Kitchen.lan (192.168.0.160)
Host is up (0.39s latency).
MAC Address: 5C:DD:DD:FF:FF:B5 (Texas Instruments)
Nmap scan report for 192.168.0.181
Host is up (0.60s latency).
MAC Address: 40:DD:DD:8F:FF:F5 (Asustek Computer)
Nmap scan report for TL-WPA4220.lan (192.168.0.225)
Host is up (0.61s latency).
MAC Address: 50:DD:FF:AA:DD:BA (Tp-link Technologies)
Nmap scan report for f3d0r4.lan (192.168.0.165)
Host is up.
Nmap done: 256 IP addresses (7 hosts up) scanned in 9.11 seconds
```

Mind you that `-sn` was known as `-sP` in the previous versions of Nmap. The use of `-sP` is still backward compatible and should work in the recent versions of Nmap.

[Free eBook: [Manage your Linux environment for success.](#)]

While Nmap man pages are well-written and provide many examples, there are specific things you won't find in the man pages. For example, what if we wanted to store IP addresses from the above output to a file? This is something specific and does not belong in the man pages of Nmap. We have to parse the output ourselves and extract IP addresses only.

For example:

```
# nmap -sn 192.168.0.0/24 | awk '/Nmap scan/{gsub(/[()]/,"",$NF); print $NF > "nmap_scanned_ips"}
```

Nmap offers many other options to save the scan output to different formats.

For example:

`-oN/-oX/-oS/-oG <file>`: Output scan in normal, XML, s|<rlpt klddi3, and Grepable format, respectively, to the given filename.

So running:

```
# nmap -sn 192.168.0.0/24 -oG nmap_output
```

produces the following output:

```
# cat nmap_output
# Nmap 7.80 scan initiated Fri Mar 6 22:01:57 2020 as: nmap -sn -oG nmap_output 192.168.0.0/24
Host: 192.168.0.1 (Archer.lan) Status: Up
Host: 192.168.0.101 (Lyric-1111C2.lan) Status: Up
Host: 192.168.0.151 (SoundTouch-VW-benee.lan) Status: Up
Host: 192.168.0.160 (SoundTouch-VW-keuken.lan) Status: Up
Host: 192.168.0.181 () Status: Up
Host: 192.168.0.225 (TL-WPA4220.lan) Status: Up
Host: 192.168.0.165 (f3d0r4.lan) Status: Up
# Nmap done at Fri Mar 6 22:02:06 2020 -- 256 IP addresses (7 hosts up) scanned in 9.45 seconds
```

 How we use cookies

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#). By using this website you agree to our use of cookies.

argets. If we were interested in checking the state of ports 22 and 443 (which 3;


```
# nmap -sV -p 22,443 192.168.0.0/24
```

If you are unsure what `-sV` does, just run:

```
# nmap | grep -- -sV
```

The above command displays the ports regardless of their state: open, closed, filtered, etc. Most of the time, we're interested in open ports, and so we can add the `-open` flag to achieve this. We'll slightly modify the above command and run:

```
# nmap -sV -p 22,443 192.168.0.0/24 -open
```

Instead of using a comma to specify a port, it is also possible to use a range of ports, which is much more flexible and easier to read. For example:

```
# nmap -p 54-111 192.168.0.0/24
```

[Cheat sheet: [Old Linux commands and their modern replacements](#)]

Advanced Nmap scanning

Now we know the basics of `Nmap` and its capabilities. Let's move to a more advanced approach to scanning targets, getting more information from a target, and using packet-tracing.

Tracing a packet on a single IP

At the moment of writing, I am connected to my server via SSH. To demonstrate how packet tracing is done using `Nmap` and what the output of such a trace looks like we are going to use the following `Nmap` syntax to produce the following output:

```
# nmap -vv -n -sn -PE -T4 --packet-trace 192.168.2.3
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-06 23:14 CET
Initiating Ping Scan at 23:14
Scanning 192.168.2.3 [1 port]
SENT (0.0282s) ICMP [192.168.0.165 > 192.168.2.3 Echo request (type=8/code=0) id=8524 seq=0] IP
[ttl=43 id=25141 iplen=28 ]
RCVD (0.0336s) ICMP [192.168.2.3 > 192.168.0.165 Echo reply (type=0/code=0) id=8524 seq=0] IP
[ttl=63 id=27840 iplen=28 ]
Completed Ping Scan at 23:14, 0.03s elapsed (1 total hosts)
Nmap scan report for 192.168.2.3
Host is up, received echo-reply ttl 63 (0.0055s latency).
Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
      Raw packets sent: 1 (28B) | Rcvd: 1 (28B)
```

The above flags have the following meanings:

- `-vv` (Increase verbosity)
- `-n` (No DNS resolution. This speeds up our scan!)
- `-sn` (No port scan)
- `-PE` (Use ICMP echo request queries. This is what is displayed in the output above)
- `-T4` (prohibits the dynamic scan delay from exceeding 10 ms for TCP ports. See `man nmap`).
- `--packet-trace` (Trace sent and received packets)

Using recursive DNS proxies for a stealth scan on a target

Let's see if we can scan on any responsive host. Let's see if we can gather some information about a target IP address. The first part is because we'll use public DNS servers, namely 8.8.4.4 and 8.8.8.8,

 **How we use cookies**

×

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#). By using this website you agree to our use of cookies.

[Work automation for everyone, a free eBook from Red Hat. \]](#)

SSH server, which results in the following:

```
# host redhat.com 8.8.8.8
Using domain server:
Name: 8.8.8.8
Address: 8.8.8.8#53
Aliases:

redhat.com has address 209.132.183.105
redhat.com mail is handled by 10 us-smtp-inbound-2.mimecast.com.
redhat.com mail is handled by 10 us-smtp-inbound-1.mimecast.com.
```

Second, let's run a stealth list scan `-sL` on the IP address 209.132.183.105.

```
# nmap --dns-servers 8.8.4.4,8.8.8.8 -sL 209.132.183.105/24
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-07 00:22 CET
Nmap scan report for network (209.132.183.0)
Nmap scan report for elvis.redhat.com (209.132.183.1)
Nmap scan report for ns2.redhat.com (209.132.183.2)
Nmap scan report for ovpn-phx2.redhat.com (209.132.183.3)
Nmap scan report for mimecast-mx01.redhat.com (209.132.183.4)
Nmap scan report for selfservice.redhat.com (209.132.183.5)
Nmap scan report for unused (209.132.183.6)
Nmap scan report for unused (209.132.183.7)
Nmap scan report for siperimeter.redhat.com (209.132.183.8)
< ----- >
< ----- >
< ----- >
```

We're able to obtain a lot of information about specific networks by using just a few simple techniques.

NSE scripts

As mentioned earlier, **Nmap** is equipped with many advanced features, one of which is NSE (Nmap Scripting Engine) scripts. Using NSE scripts with **Nmap** allows you to scan different hosts and find vulnerabilities in services running on the host and possibly log in by brute-forcing these services.

The use of NSE script syntax is as follows:

```
# nmap --script="name_of_script" --script-args="argument=arg" target
```

Now, you are probably wondering where to find these NSE scripts and how to know what script uses what arguments. Start by running `man nmap`. You can also jump straight away to the right section, i.e.:

```
# PAGER='less "+/NMAP SCRIPTING ENGINE"' man nmap
```

The available NSE scripts you can pass to **Nmap** are located at:

`/usr/share/nmap/scripts/`

You can also locate the NSE scripts by running:

```
# dnf -y install mlocate ; updatedb ; locate nmap/scripts
```

Now that we know where NSE scripts are located let's see how we can use these scripts to get some information about a target that's running a web server.

How we use cookies

×

bsite

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#). By using this website you agree to our use of cookies.

ed to protect websites from SQL injection, cross-site scripting, malformed site is protected by such a WAF. The following displays the usage of an NSE

```
# nmap -p443 --script http-waf-detect --script-args="http-waf-detect.aggro,http-waf-detect.detectBodyChanges" www.slimmer.ai
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-09 22:38 CET
Nmap scan report for www.slimmer.ai (172.104.131.188)
Host is up (0.023s latency).
rDNS record for 172.104.131.188: li1647-188.members.linode.com

PORT      STATE SERVICE
443/tcp   open  https
| http-waf-detect: IDS/IPS/WAF detected:
|_www.slimmer.ai:443/?p4yl04d=id;uname%20-a

Nmap done: 1 IP address (1 host up) scanned in 1.37 seconds
```

As shown above, a Web Application Firewall protects the target website.

More NSE scripts

Once again, Nmap is often used by system administrators to inventory their environment, discover weaknesses in their network, and so protect their systems from intruders. Intruders, on the other hand, can do the same to explore a remote system and try to gather as much information as possible about such a system.

Assume that some unauthorized person has scanned your network and found a few open ports/services. This person could then pass some NSE scripts to Nmap and see if these services are vulnerable. Here is what is going to happen:

```
# nmap -Pn -sV --script=vulners 37.xx.xx.xx
Starting Nmap 7.80 ( https://nmap.org ) at 2020-03-09 22:41 CET
Nmap scan report for some.domain.nl (37.xx.xx.xx)
Host is up (0.016s latency).
Not shown: 998 filtered ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.4 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:7.4:
|       CVE-2018-15919  5.0   https://vulners.com/cve/CVE-2018-15919
|_       CVE-2017-15906  5.0   https://vulners.com/cve/CVE-2017-15906
25/tcp    open  smtp      Postfix smtpd
Service Info: Host: some.domain.nlService detection performed. Please report any incorrect results
at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 18.20 seconds
```

We can see that the remote system is running OpenSSH 7.4. Nmap queried public vulnerability databases and found the known CVE's.

Wrap up

Nmap is a very powerful system inventory and port scanning tool that can be used for good and bad purposes. It depends on which hat you are wearing. The best way to learn Nmap is to read man pages, use examples shown in the man pages, and experiment with the NSE scripts. Also, try Zenmap. If you are interested in knowing more about port scanning and the science behind it, see the [Nmap documentation](#).

Topics:

Networking

Linux

Troubleshooting

How we use cookies

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#).

By using this website you agree to our use of cookies.



Valentin Bajrami

Valentin is a system engineer with more than six years of experience in networking, storage, high-performing clusters, and automation. He is involved in different open source projects like bash, Fedora, Ceph, FreeBSD and is a member of Red Hat Accelerators. [More about me](#)

Try Red Hat Enterprise Linux

Download it at no charge from the Red Hat Developer program.

Related Content



[Test TCP ports with Python and Scapy](#)

Get greater control over TCP port checking with a DIY, customizable approach using Python and Scapy.

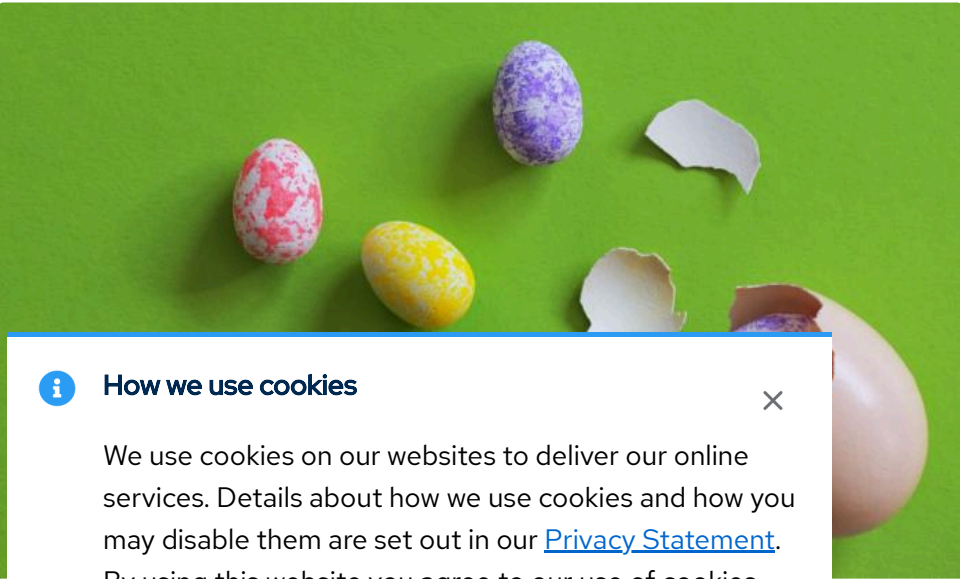
Posted: April 19, 2023
Author: [Jose Vicente Nunez](#) (Sudoer).



[Stop using Telnet to test ports](#)

Make life simpler by automating network checks with tools like Expect, Bash, Netcat, and Nmap instead.

Posted: April 18, 2023
Author: [Jose Vicente Nunez](#) (Sudoer).



How we use cookies

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#).
By using this website you agree to our use of cookies.

[8 open source 'Easter eggs' to have fun with your Linux terminal](#)

Hunt these 8 hidden or surprising features to make your Linux experience more entertaining.

Posted: April 10, 2023
Author: [Ricardo Gerardi](#) (Editorial Team, Sudoer alumni, Red Hat).

The opinions expressed on this website are those of each author, not of the author's employer or of Red Hat. The content published on this site are community contributions and are for informational purpose only AND ARE NOT, AND ARE NOT INTENDED TO BE, RED HAT DOCUMENTATION, SUPPORT, OR ADVICE.

Red Hat and the Red Hat logo are trademarks of Red Hat, Inc., registered in the United States and other countries.



- About Red Hat
- Jobs
- Events
- Locations
- Contact Red Hat
- Red Hat Blog
- Diversity, equity, and inclusion
- Cool Stuff Store
- Red Hat Summit

-
- © 2023 Red Hat, Inc.
 - Privacy statement
 - Terms of use
 - All policies and guidelines
 - Digital accessibility
 - Cookie preferences



How we use cookies

×

We use cookies on our websites to deliver our online services. Details about how we use cookies and how you may disable them are set out in our [Privacy Statement](#). By using this website you agree to our use of cookies.